THE UNIVERSITY OF THE SOUTH PACIFIC

School of Information Technology, Engineering, Mathematics, and Physics

CS311: Operating Systems

Short Test 1 Semester 1, 2025

Mode: Face to Face

Duration of Exam: 1 Hour 5 minutes

Reading Time: Extra 2 minutes

Writing Time: 1 Hour 5 minutes

Instructions:

All questions are compulsory. Weighting: 15% | Total Marks: 50

Calculators are allowed. This is a closed book exam.

Learning Outcomes

I. Solve synchronization problems inherent in inter-process communication and multi-threaded applications.

II. Evaluate various techniques for efficient management of operating system resources.

Student ID: _____

Student Name: _____

Question 1 **15 Marks**

Consider the following snapshot of a system in which four resources A, B, C and D are available. The system contains a total of 2 instances of resource A, 2 instances of resource B, 1 instance of resource C and 1 instance of resource D.

	Allocation				**Maximum**				**Available**			
Process	A	B	C	D	A	B	C	D	A	B	C	D
P1	1	0	0	0	3	3	1	1	2	2	1	1
P2	0	1	0	0	1	1	0	2				
P3	1	0	1	0	3	2	1	0				
P4	0	0	0	1	3	1	1	1				

a) Compute the need matrix for the processes in the system. **3 Marks**

b) Is the system in a safe state or is it deadlocked? Justify your answer. **5 Marks**

c) If a request from P3 arrives for (2, 1, 0, 0), can the request be granted immediately? Perform a safety test to determine the execution order and check for deadlocks. **7 Marks**

Question 2 **6 Marks**

Use the program shown below to answer the following questions:

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>
    int value = 5;
    int main()
   {
        pid_t pid;

        pid = fork();

    if (pid == 0) { /* child process */
        value += 15;
        return 0; }

    else if (pid > 0) { /* parent process */
    wait(NULL);
    printf("PARENT: value = %d",value); /* LINE A */
    return 0;  }
}
```

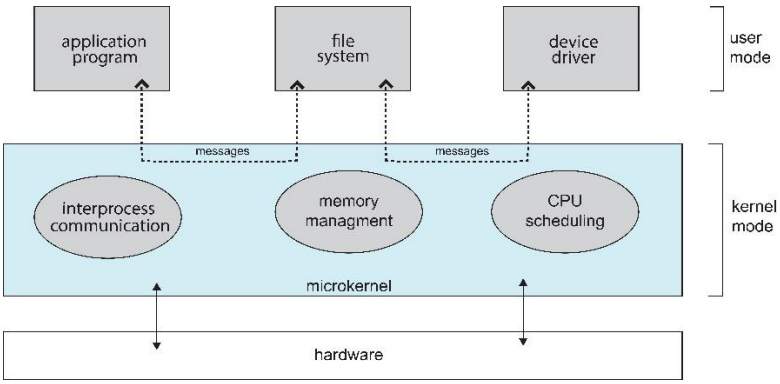
a) Explain the purpose of fork () and wait(NULL) in the program. **2 Marks**

a) Analyse the memory space of the parent and child process and provide the values stored in each. **2 Marks**

b) Determine the output at Line A and the number of processes created. Justify your answer. **2 Marks**

Question 3 **7 Marks**

Use the diagram below to answer the following questions:



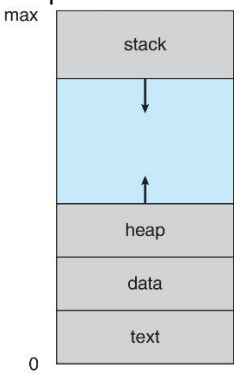
a) Provide reasons why memory management is considered a privileged instruction. **2 Marks**

b) Explain the design illustrated in the diagram above and state the advantage it has over monolithic design. **2 ½ Marks**

c) Describe the models used in interprocess communication and state the three general methods for passing parameters to the operating system. **2 ½ Marks**

Question 4 **8 Marks**

a) Outline the importance of the heap section of a process and what would happen if the stack and heap collide? **2 Marks**



b) Illustrate the diagram of a process state and clearly label the trigger points. **2 Marks**

c) Compare and contrast POSIX , Windows 32 and Grand Central Dispatch (GCD) system calls. Differentiate in terms of architecture and functionality. **4 Marks**

Question 5 **10 Marks**

Consider the following set of processes :

Process	Priority ID	Burst Time (ms)	Arrival Time (ms)
P1	1	7	2
P2	1	6	0
P3	2	4	1
P4	2	2	4

a) Assume the first 3 milliseconds is left idle after which pre-emptive priority scheduling is applied. Draw the Gantt chart and compute the CPU utilization rate. For processes with the same priority, use a time quantum of 2 milliseconds. **4 Marks**

i. Based on the arrival times indicated in the table, draw a Gantt chart using pre-emptive SJF scheduling. **4 Marks**

ii. Compute the average completion time for the processes assuming non-pre-emptive FCFS scheduling.

2 Marks

Question 6 **4 Marks**

Use the program below to answer the question:

export OMP_NUM_THREADS=2

```c
#include <omp.h>
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char* argv[])
{
    #pragma omp parallel
    {
        printf("All the best from Thread ID = %d\n",
            omp_get_thread_num());
    }
}
```

a) Write the commands to compile the program in a Linux environment given the filename as test.c

1 Mark

b) Write the configurations to execute the file once debugging is complete with the specified threads as provided in the program and display the output. **3 Marks**

THE END 😊